

1. Project Overview

Project Name: Automatic Task Scheduler

Purpose: Manage tasks automatically using date-time scheduling. Generate desktop popup reminders and email notifications. Store task data in CSV and track completed, pending, and failed tasks.

2. Test Objectives

Verify that:

1. User login and new registration works correctly.
2. Tasks are added, updated, deleted, and viewed correctly.
3. Task completion moves data to completed.csv.
4. Desktop popup notification is triggered at scheduled time.
5. Email is sent 10 minutes before and after missed task.
6. Tasks are sorted by status and prioritized by date-time.

7. Dashboard displays correct statistics.
8. Invalid inputs and errors are handled correctly.
9. No data loss occurs in CSV files.

3. Modules Under Test

Module	Priority
User Authentication	High
Task-Management-CRUD	High
Task Completion	High
Scheduler-Popup Notification	High
Email Notification	High
Sort & Prioritize	High
Dashboard	Medium
CSV Storage	High

4. Passes Test Cases

Test Case ID	Test Scenario	Excepted Result	Status	Priority
TC-01	Valid user login	Login successful	Pass	High
TC-02	Invalid password	Invalid password message	Pass	High
TC-03	New user registration	User created in user.csv	Pass	High
TC-04	Add new task with valid data	Task add in Tasks.csv	Pass	High
TC-05	Add task with empty field	Error: All fields required	Pass	High
TC-06	Add task with invalid date	Invalid date format error	Pass	High
TC-07	Update existing task	Task updated successfully	Pass	High
TC-08	Delete existing task	Task deleted successfully	Pass	High
TC-09	Undo delete task	Task restored successfully	Pass	High
TC-10	View all tasks	All tasks displayed	Pass	Medium
TC-11	Complete valid task	Task moved to completed.csv	Pass	High

TC-12	Complete invalid task	Task ID not found message	Pass	High
TC-13	Desktop popup at scheduled time	Popup notification displayed	Pass	High
TC-14	10-min email reminder	Email sent before task	Pass	High
TC-15	Missed task email	Failure email sent	Pass	High
TC-16	Sort tasks by status	3 files created:completed,pending,failed	Pass	High
TC-17	Prioritize tasks by datetime	prioritized.csv created	Pass	High
TC-18	Dashboard statistics	Correct count of all task types displayed	Pass	Medium

5. Bugs Found

Bug No	Test Scenario	Issue	Status	Priority
BUG-001	Email notification	Email only to hardcoded gmail,not user email	Fail	High
BUG-002	Email validation	No validation for email format in email_notification.py	Fail	High

BUG-003	Duplicate Task_ID	System allows multiple task with same Task_ID	Fail	High
BUG-004	CSV file missing	If Tasks.csv deleted,program gives error	Fail	High
BUG-005	Date format validation	User can enter wrong date format if not validated	Fail	High
BUG-006	Scheduler dependency	Scheduler must be kept running manually for notifications	Fail	High
BUG-007	Undo delete limitation	Only last deleted task can be restored,not multiple	Fail	Medium
BUG-008	Dashboard logic	Failed tasks counted only if time <current time	Fail	Medium
BUG-009	Email credentials	Gmail app password required. if wrong email fails	Fail	High
BUG-010	Recurring task	Recurring tasks not auto-generated after completion	Fail	High
BUG-011	CSV storage path	User don't know where the CSV files are saved	Fail	Medium
BUG-012	Input validation	User can enter every long task names. No	Fail	High

		validation for inputs		
BUG-013	Task overwrite	When new task same id added,old task gets overwritte	Fail	High
BUG-014	Invalid uantity/price	Application crashes when non-numeric values entered	Fail	High
BUG-015	Cancel option	User cannot cancel task creation once started	Fail	Medium

6 .Functional Test Cases

TC ID	Test Scenario	Test steps	Expected Result
01	Verify task creation	Open scheduler > Create new task > Set file/folder path + schedule time/frequency	Task is created and appears in task list with correct details
02	Verify scheduled execution	Create task to run at specific time > Wait for scheduled time	Task executes automatically at the scheduled time without manual intervention

03	Verify recurring task	Create daily/weekly/monthly recurring task	Task runs automatically at each defined interval
04	Verify task pause/resume	Pause an active scheduled task > Resume it later	Task pauses successfully and resumes execution from next scheduled run
05	Verify task deletion	Delete an existing scheduled task from task list	Task is removed and no longer executes at scheduled time
06	Verify invalid path handling	Create task with non-existent file/folder path	System shows error "Invalid path" and task does not run
07	Verify task log maintenance	Run a task and check scheduler logs	Logs show task name, execution time, status, and result
08	Verify system	Create task>Restart system>Wait for next	Task resumes

	restart behavior	schedule	scheduling automatically after system reboot
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